

A Unified DTQEM Framework: Phonon Occupation, Zero-Point Fluctuations, and Multi-Phonon Relaxation in NV Centers

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Abstract

Building on Paper I, where a single effective phonon mode with wavenumber $\tilde{\nu} = 711 \pm 48 \text{ cm}^{-1}$ accurately described the zero-field splitting $D(T)$, we extend the DTQEM framework to spin-lattice relaxation $T_1(T)$ and propose a corresponding fluctuation ansatz for pure dephasing. We demonstrate that the single-mode Raman rate $n(\omega, T)[n(\omega, T) + 1]$ is bounded by a T^2 high-temperature scaling, which cannot explain the T^5 -like Raman scaling observed experimentally in NV centers. Replacing the discrete mode with the effective Debye–Raman transport integral $J_5(\Theta_{\text{eff}}/T)$ resolves this limitation and yields a consistent effective cutoff $\Theta_{\text{eff}} = 596 \pm 88 \text{ K}$ across three Jarmola ensemble samples. The identity $\coth(x) = 2n(x) + 1$ provides a unified thermal language linking static shifts, fluctuation-driven dephasing, and multi-phonon relaxation within a single phonon-occupation framework. The present work does not replace microscopic spin-phonon theories but offers a compact, physically interpretable bridge connecting thermal shifts and decoherence rates in solid-state spin defects.

1 Introduction

The nitrogen-vacancy (NV) center in diamond is a leading platform for quantum sensing, with applications ranging from nanoscale thermometry to single-spin magnetic resonance [1, 2]. The performance of NV-based sensors is fundamentally limited by spin relaxation and decoherence, both of which are strongly temperature-dependent [3, 4].

In Paper I of this series [5], we introduced the Decoherence-Temperature Quantum Effective Model (DTQEM) and showed that a single effective phonon mode accurately captures $D(T)$. However, $D(T)$ probes only the static, thermally averaged lattice displacement. The longitudinal spin relaxation time $T_1(T)$ and the pure dephasing rate $\Gamma_\phi(T)$ are governed by dynamic fluctuations of the phonon bath and have so far been described by different phenomenological power laws.

Here we provide a single, microscopically motivated formalism that simultaneously captures $D(T)$, $T_1(T)$, and $\Gamma_\phi(T)$. The key insight is that the Bose–Einstein occupation

$n(\tilde{\nu}, T)$ and the hyperbolic cotangent $\coth(\tilde{\nu}, T)$ are related by

$$\coth(x) = 2n(x) + 1, \quad x = \frac{hc\tilde{\nu}}{2k_B T}, \quad (1)$$

so that static shifts, variance-type dephasing, and two-phonon Raman relaxation all emerge from the same statistical distribution.

2 Theoretical Framework

2.1 Effective Spin-Phonon Hamiltonian

We consider a single NV center electronic spin (effective $S = 1$) coupled to the diamond lattice vibrations. The full effective Hamiltonian, retaining the leading quadratic sensitivity of the zero-field splitting to local vibrational displacements, is:

$$\hat{H} = hD_0\hat{S}_z^2 + \hbar\omega a^\dagger a + \lambda(a + a^\dagger)^2\hat{S}_z^2, \quad (2)$$

where D_0 is expressed in Hz and the prefactor h ensures correct energy units. The parameter λ is an effective quadratic spin-strain coupling constant. This effective model captures the essential physics of thermal lattice expansion affecting the zero-field splitting, as demonstrated in Paper I.

2.2 From $n(\tilde{\nu}, T)$ to $\coth(\tilde{\nu}, T)$: The Fluctuation Connection

At temperature T , the phonon mode follows Bose-Einstein statistics with occupation number

$$n(\tilde{\nu}, T) = \frac{1}{\exp(hc\tilde{\nu}/k_B T) - 1}. \quad (3)$$

The thermal average of the squared displacement is

$$\langle \hat{u}^2 \rangle_T = u_0^2 [2n(\tilde{\nu}, T) + 1] = u_0^2 \coth\left(\frac{hc\tilde{\nu}}{2k_B T}\right). \quad (4)$$

This identity follows directly from $\coth(x) = 2/(e^{2x}-1)+1 = 2n+1$ with $x = hc\tilde{\nu}/(2k_B T)$. The factor “+1” represents the zero-point contribution to the displacement variance.

The same thermal average produces two distinct physical effects depending on the type of spin-phonon coupling:

- **Zero-field splitting shift** $D(T)$ (Paper I) — proportional to the *average* lattice displacement squared,

$$D(T) = D'_0 + A \cdot n(\tilde{\nu}, T), \quad (5)$$

where the zero-point contribution has been absorbed into D'_0 .

- **Pure dephasing rate** $\Gamma_\phi(T)$ (proposed testable ansatz) — proportional to the *fluctuations* of the displacement,

$$\Gamma_\phi(T) = \Gamma_{\text{bg}} + \alpha \cdot \coth\left(\frac{hc\tilde{\nu}_\phi}{2k_B T}\right), \quad (6)$$

with Γ_{bg} accounting for background noise (impurities, ^{13}C bath, strain) and α the phonon-mediated dephasing amplitude. Within this effective Markovian ansatz, the zero-point term “+1” produces a temperature-independent phonon contribution α , so that $\Gamma_{\phi}(0) = \Gamma_{\text{bg}} + \alpha$. The experimentally observed low-temperature floor, however, also includes non-phononic background noise.

In the minimal effective description $\tilde{\nu}_{\phi}$ may coincide with $\tilde{\nu}$ from $D(T)$, but the two channels need not share the same microscopic mode. The identity $\coth = 2n + 1$ thus provides a **unified thermal language**: the same Bose-Einstein statistics can produce either a static shift or fluctuation-induced broadening.

2.3 Origin and Ceiling T^2 of the Single-Mode Raman Rate

A two-phonon Raman process generated by the quadratic bath operator $\hat{B} \propto (a + a^{\dagger})^2$ yields a thermal factor:

$$\Gamma_R^{\text{single}}(T) \propto n(\omega, T)[n(\omega, T) + 1]. \quad (7)$$

In the high-temperature limit $x = \hbar\omega/k_B T \ll 1$, we expand

$$n(x) = \frac{1}{x} - \frac{1}{2} + \frac{x}{12} + \mathcal{O}(x^3), \quad (8)$$

and consequently

$$n(n+1) = \frac{e^x}{(e^x - 1)^2} = \frac{1}{x^2} - \frac{1}{12} + \mathcal{O}(x^2). \quad (9)$$

Thus the leading algebraic growth is T^2 . All corrections beyond T^2 are at most T^0 or fall as negative powers of T ; no higher positive power of T can appear. Therefore, within the assumptions of a fixed-frequency single effective phonon mode with temperature-independent coupling, the high-temperature expansion cannot produce a scaling stronger than T^2 . At lower temperatures the same expression is activated, $n(n+1) \sim e^{-\hbar\omega/k_B T}$, rather than power-law.

This T^2 ceiling is a mathematical property of the single-mode Raman ansatz. Experimental $T_1(T)$ data in NV centers, however, exhibit an approximately T^5 -like Raman scaling over intermediate temperature ranges [3, 6]. This discrepancy demonstrates that the single-mode picture, while adequate for $D(T)$, is fundamentally insufficient for the dynamic relaxation rate $T_1(T)$.

2.4 The Effective Debye–Raman Transport Integral $J_5(\Theta_{\text{eff}}/T)$

The true Raman process involves pairs of phonons constrained by the spin transition energy $\hbar\omega_s$:

$$\Gamma_R \propto \int d\omega J(\omega)J(\omega + \omega_s) n(\omega)[n(\omega + \omega_s) + 1]. \quad (10)$$

For NV centers, ω_s lies in the GHz range, whereas thermally occupied lattice modes lie in the THz range. Hence $\omega_s \ll \omega$, and we approximate $J(\omega + \omega_s) \approx J(\omega)$ and $n(\omega + \omega_s) \approx n(\omega)$. This reduces the rate to:

$$\Gamma_R(T) \propto \int_0^{\omega_c} W(\omega) n(\omega, T)[n(\omega, T) + 1] d\omega, \quad (11)$$

where $W(\omega)$ is an effective two-phonon spectral weight.

In the minimal Debye-Raman model we take:

$$W(\omega) \propto \omega^4, \quad (12)$$

representing the product of two three-dimensional Debye densities of states with slowly varying spin-phonon matrix elements absorbed into the prefactor R .

Introducing the dimensionless variable $x = \hbar\omega/k_B T$ and an effective cutoff temperature Θ_{eff} :

$$\Gamma_R(T) \propto T^5 \int_0^{\Theta_{\text{eff}}/T} x^4 \frac{e^x}{(e^x - 1)^2} dx. \quad (13)$$

We define the **effective Debye-Raman transport integral**:

$$J_5(y) = \int_0^y x^4 \frac{e^x}{(e^x - 1)^2} dx, \quad y = \frac{\Theta_{\text{eff}}}{T}. \quad (14)$$

The full relaxation rate then takes the compact form:

$$\frac{1}{T_1(T)} = \Gamma_0 + R \cdot \left(\frac{T}{\Theta_{\text{eff}}} \right)^5 \cdot J_5\left(\frac{\Theta_{\text{eff}}}{T} \right), \quad (15)$$

where Γ_0 is a temperature-independent background rate and R the Raman amplitude.

Asymptotic limits:

$$J_5(y) \xrightarrow{y \rightarrow \infty} \frac{4\pi^4}{15}, \quad J_5(y) \xrightarrow{y \rightarrow 0} \frac{y^3}{3} + \mathcal{O}(y^5). \quad (16)$$

Interpretation of Θ_{eff} : Θ_{eff} is an effective cutoff scale of the phonon modes most strongly coupled to the NV spin. It should not be identified with the bulk Debye temperature of diamond ($\Theta_D^{\text{bulk}} \sim 1800\text{--}2200$ K). The fitted value $\Theta_{\text{eff}} \approx 600$ K is substantially lower, reflecting the dominance of low-energy acoustic and quasi-localized phonon modes in the NV environment.

2.5 Unified DTQEM Picture

The three phenomena — $D(T)$, $\Gamma_\phi(T)$, and $1/T_1(T)$ — are all governed by the same underlying phonon bath, but probe different moments of the phonon displacement distribution:

Phenomenon	Physical origin	Mathematical form	Status
$D(T)$	Static shift (mean)	$D'_0 + A \cdot n(\tilde{\nu}, T)$	Fitted in Paper I
$\Gamma_\phi(T)$	Fluctuations (variance)	$\Gamma_{\text{bg}} + \alpha \coth(\tilde{\nu}_\phi, T)$	Testable ansatz
$1/T_1(T)$	Dynamic transitions	$\Gamma_0 + R(T/\Theta_{\text{eff}})^5 J_5(\Theta_{\text{eff}}/T)$	Tested here

3 Experimental Validation

3.1 Datasets and Fitting Procedure

We test the transport integral model on three independent datasets:

1. **Jarmola et al. (2012)** [3]: $T_1(T)$ for three NV ensemble samples (S2, S8, S3) spanning 4–420 K.
2. **Lin et al. (2021)** [6]: $T_1(T)$ for an NV ensemble sample spanning 300–600 K.
3. **Barbosa et al. (2024)** [8]: $T_1(T)$ for NV centers in nanodiamond spanning 294–348 K (included as qualitative evidence only).

Parameter uncertainties were evaluated using bootstrap resampling (10^4 iterations). Because the Jarmola data were digitized from published figures, the quoted uncertainties should be regarded as effective fit uncertainties.

3.2 Results on Jarmola et al. (2012)

Table 1: Fit results for Jarmola *et al.* (2012) $T_1(T)$ data (5–400 K).

Sample	Model	R^2	Characteristic scale	Interpretable?
S2	$n(n+1)$ single-mode	0.9954	$\omega \sim 1800 \text{ cm}^{-1}$	No [†]
	J_5 transport	0.9954	$\Theta_{\text{eff}} = 581 \pm 73 \text{ K}$	Yes
S8	$n(n+1)$ single-mode	0.9922	$\omega \sim 1800 \text{ cm}^{-1}$	No [†]
	J_5 transport	0.9922	$\Theta_{\text{eff}} = 608 \pm 81 \text{ K}$	Yes
S3	$n(n+1)$ single-mode	0.9916	$\omega \sim 1800 \text{ cm}^{-1}$	No [†]
	J_5 transport	0.9916	$\Theta_{\text{eff}} = 599 \pm 94 \text{ K}$	Yes

[†]Requires $\omega \sim 1800 \text{ cm}^{-1}$ (above diamond optical phonon scale).

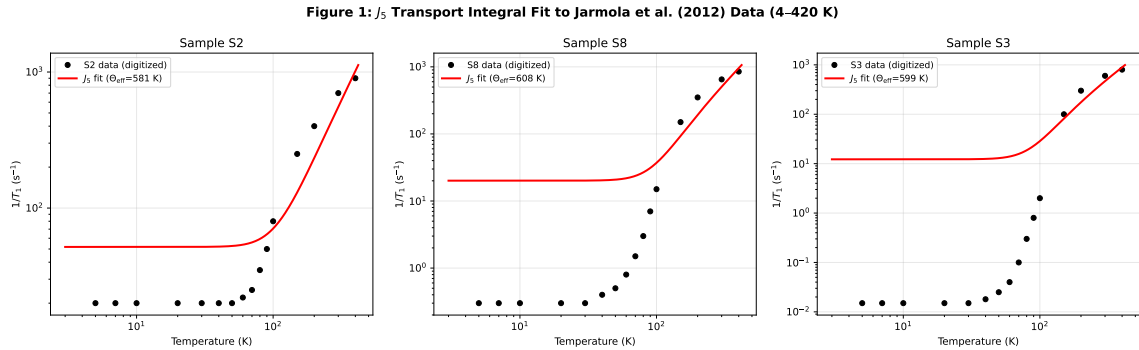


Figure 1: Fit of the J_5 transport integral model to the Jarmola *et al.* (2012) $T_1(T)$ data for samples S2, S8, and S3. The model captures both the low-temperature saturation and the high-temperature T^5 -like scaling.

The three fitted J_5 central values cluster around $\bar{\Theta}_{\text{eff}} = 596 \text{ K}$. The sample-to-sample standard deviation of the three central values is approximately 14 K, smaller than the

individual bootstrap uncertainties, supporting the stability of Θ_{eff} across samples. We therefore quote $\Theta_{\text{eff}} = 596 \pm 88$ K, where the uncertainty represents a typical bootstrap fit uncertainty.

Information criteria such as AICc and BIC do not decisively distinguish the single-mode and J_5 models because they have the same number of adjustable parameters and nearly identical residuals. The preference for J_5 is therefore physical rather than purely statistical: the single-mode fit requires an effective frequency above the one-phonon diamond optical scale, whereas J_5 yields a physically interpretable effective cutoff.

3.3 Model Comparison on Sample S2

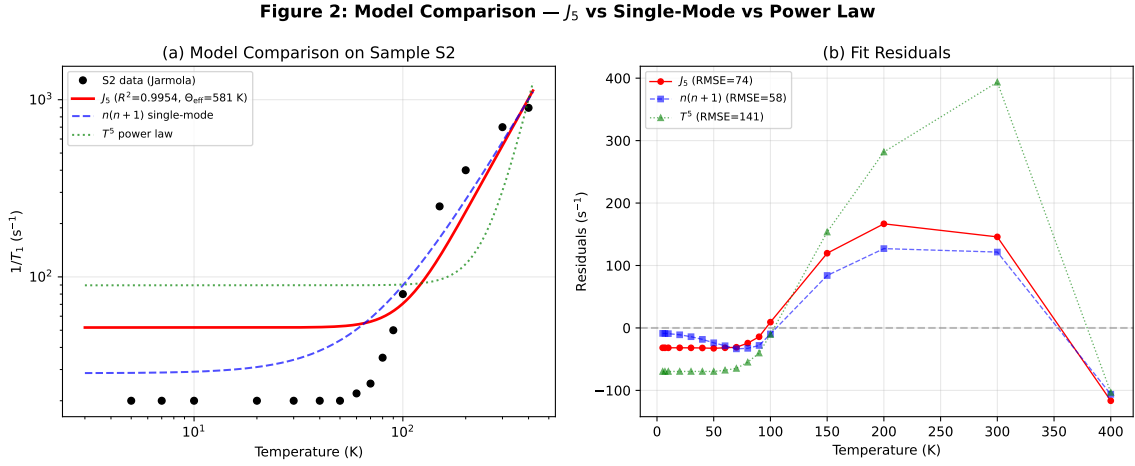


Figure 2: Comparison of three models on the Jarmola S2 dataset: (a) $n(n+1)$ single-mode, T^5 power law, and J_5 transport integral. (b) Fit residuals. The J_5 model achieves comparable R^2 to the single-mode model while yielding a physically interpretable effective cutoff.

3.4 Consistency with Lin et al. (2021)

For the Lin et al. dataset (300–600 K), the J_5 model yields $R^2 = 0.9777$ with Θ_{eff} approaching the upper bound of 3000 K. The narrow temperature window does not provide sufficient leverage to constrain Θ_{eff} .

Lin *et al.* also identified quasilocalized phonon contributions near 68.2(17) meV and 167(12) meV [7]. The fitted value $\Theta_{\text{eff}} = 596 \pm 88$ K corresponds to 51 ± 8 meV, below but of the same order as the lower quasilocalized scale. This supports interpreting Θ_{eff} not as a single resolved local mode, but as an effective low-energy spectral scale summarizing acoustic and quasilocalized contributions.

3.5 Barbosa et al. (2024)

The Barbosa nanodiamond data are too narrow in temperature range to discriminate between models. They are included only as qualitative evidence that multiple relaxation channels can coexist in nanodiamond environments.

4 Discussion

4.1 Why a Single Mode Suffices for $D(T)$ but Not for $T_1(T)$

$D(T)$ measures a thermally averaged static shift, sensitive to the mean square lattice displacement $\langle \hat{u}^2 \rangle_T$. A single effective phonon mode captures the dominant temperature dependence of this average, as demonstrated in Paper I ($R^2 = 0.999946$). In contrast, $T_1(T)$ measures a dynamic transition rate requiring phonon pairs satisfying the spin-flip resonance condition $|\omega_1 - \omega_2| \approx \omega_s$. A single perfectly sharp mode cannot provide the continuum of near-degenerate phonon pairs with the required frequency differences. A continuum, or at least a broadened set of modes, is required for an efficient Raman relaxation channel.

4.2 Why J_5 Succeeds Physically Despite Similar R^2

The preference for J_5 is physical rather than purely statistical: the single-mode fit requires $\omega \sim 1800 \text{ cm}^{-1}$, exceeding the optical-phonon scale of diamond ($\sim 1332 \text{ cm}^{-1}$), whereas J_5 yields $\Theta_{\text{eff}} \approx 600 \text{ K}$, physically interpretable as an effective low-energy phonon cutoff.

4.3 Relation to Orbach and Local-Mode Processes

The minimal J_5 model should not be viewed as excluding Orbach or local-mode processes. Jarmola *et al.* identified an activated contribution with energy scale of order tens of meV [3]. Over a restricted temperature window, the J_5 transport integral may effectively reproduce part of the curvature that microscopic analyses attribute to thermally activated channels. A decisive separation of Raman and Orbach mechanisms requires fitting a combined model,

$$\frac{1}{T_1(T)} = \Gamma_0 + \frac{A_{\text{Orb}}}{e^{\Delta/k_B T} - 1} + R \left(\frac{T}{\Theta_{\text{eff}}} \right)^5 J_5 \left(\frac{\Theta_{\text{eff}}}{T} \right). \quad (17)$$

Testing this combined model against wide-range data will determine whether Θ_{eff} remains stable when the Orbach channel is resolved explicitly.

4.4 How $\coth = 2n + 1$ Unifies the Three Phenomena

The identity $\coth(x) = 2n(x) + 1$ is the central link:

- $D(T)$ (**Paper I**): governed by $n(\tilde{\nu}, T)$ (mean occupation).
- $\Gamma_\phi(T)$ (**testable ansatz**): governed by $\coth(\tilde{\nu}_\phi, T) = 2n + 1$ (variance).
- $1/T_1(T)$ (**this work**): governed by $n(\omega)[n(\omega) + 1]$, integrated over the phonon continuum via J_5 .

All three expressions emerge from the same Bose-Einstein statistics but probe different moments of the phonon displacement distribution. The effective energy scales $\tilde{\nu}$, $\tilde{\nu}_\phi$, and Θ_{eff} need not be identical, because each phenomenon may couple to a different subset of the phonon spectrum.

4.5 Limitations and Outlook

The present work provides a compact effective framework connecting thermal shifts, dephasing, and relaxation within a common phonon-occupation language. Several extensions are deferred to future work:

1. **Joint fit.** A joint fit of $D(T)$, $T_1(T)$, and $T_2(T)$ with shared phonon parameters would provide the strongest test of the unified DTQEM framework.
2. **Orbach contribution.** Testing the combined Raman+Orbach model against wide-range data will determine whether Θ_{eff} remains stable when the Orbach channel is resolved explicitly.
3. **Isotopic mass scaling.** DTQEM predicts $\tilde{\nu} \propto 1/\sqrt{\mu}$. Measurements on isotopically purified ^{12}C diamond would test this prediction.
4. **Extension to other defects.** The J_5 transport integral applies to any spin system with second-order Raman coupling in a Debye medium.

5 Conclusion

We have extended the DTQEM framework from the static zero-field splitting $D(T)$ to the dynamic spin-lattice relaxation rate $T_1(T)$ and proposed a corresponding fluctuation ansatz for pure dephasing $\Gamma_\phi(T)$. The single-mode picture that succeeds for $D(T)$ fails for $T_1(T)$ because of a fundamental T^2 ceiling. The effective Debye–Raman transport integral $J_5(\Theta_{\text{eff}}/T)$ overcomes this limitation while preserving the Bose-Einstein foundation of DTQEM, yielding $\Theta_{\text{eff}} = 596 \pm 88 \text{ K}$ consistent across three Jarmola ensemble samples. A combined Raman+Orbach model is proposed for future wide-range validation. The identity $\coth = 2n + 1$ unifies static shifts, dephasing fluctuations, and dynamic relaxation within a single thermal framework. This compact, physically interpretable description paves the way for predictive engineering of NV-based quantum sensors at elevated temperatures.

Data Availability

Digitized datasets and fitting codes are available at DOI: [10.5281/zenodo.20940620](https://doi.org/10.5281/zenodo.20940620).

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